CS 5493/4243 Large Scale Database Management Systems, Fall 2026

Course Project: A database application with a Web front-end

Part 1: available at 6:00 pm CST on Thursday, October 8, 2026

Due at 11:59 pm CST on Wednesday, October 21, 2026

Part 2: available at 6:00 pm CST on Thursday, October 8, 2026

Due at 11:59 pm CST on Thursday, December 3, 2026

General Information

This assignment is worth ***32%*** *(part 1 for 7%, part 2 for 20%, and 5% for presentations) of the overall grade* and is roughly equivalent to 3 assignments in terms of workload. Do not wait until the last day to get started on the project. You will not be able to finish the project one or two days before the deadline.

In this assignment you will build a database application of your choice. In part 1 you will develop a conceptual design of your application with an ER model and translate the ER model to a relational schema. We will give you feedback on part 1, which you should incorporate when working on part 2 of the project. In part 2 you will make any necessary adjustments, populate your database schema with data, and implement a simple Web application that queries and updates your database using the python package *streamlit* (https://streamlit.io/). We will use *PostgreSQL* (https://www.postgresql.org/) database for the project. Non-SQL application is possible but require special approval.

This assignment is to be completed in teams of two for undergraduate section and of team one for graduate section. Consult the course syllabus for a description of our academic honesty policy. Both partners are expected to equally participate in all parts of the project. You may discuss all aspects of the project with your partner.

Action Items and Deliverables

1. **Part 1: by October 21 @ 11:59pm CST,** submit a document with the description of your application, an ER diagram, and a schema.sql file that contains the translation of your ER diagram to the relational model.

2. **Part 2, by December 3 @ 11:59pm CST**, submit your final project report (no more than 4 pages), data (if accessible), code (python files, insertion.sql for loading data, README for execution guidelines), presentation (slides in PDF), and an individual contribution statement.

   Each project partner should prepare a statement of contributions: a document describing their own and their partner's contributions to the project. **This part of the assignment is to be completed individually.** Do not discuss the content of this document with your partner, and do not collaborate on this document.

Part 2 details

- Modify your project description, ER diagram, and schema.sql, incorporating any feedback you received from us.
- Implement a data loading procedure in an insertion.sql file. This procedure can be as simple as including insert statements. You may also use the psql copy command to load data from csv into your tables, see Documentation: 16: COPY for additional information. Finally, you may write custom code to populate the tables in your database tables with data. Whatever option you choose, you should load at least *100 tuples* of real or realistic data into each table in your database schema, keeping in mind that the queries you are implementing should give interesting results. Carefully describe your data loading procedure in the project report.
- Carefully think about the interactions the user will have with your application and implement them as part of the Web front-end. Your Web interface does not need to be fancy. However, **the user should not be asked to type in SQL queries**. You might receive extra points based on the functionality your interface supports based on the peer evaluation after the oral presentation of the project.
- Most interactions should involve some sort of input values. Input values may be specified using menus, radio buttons, checkboxes, scrollers, etc. Text input boxes may also be appropriate. As a result of the user's interaction, you should be generating a query that is executed against your database schema and displaying results at the next step.

- You are required to implement at least 5 kinds of user interactions that lead to executing a query with a join, a group by, or both. You may decide to implement additional interactions that select data from 1 table at a time, or that insert, update or modify data. Still, **at least 5 of the queries you implement must involve a join, a group by, or both**.
- You will implement the Web front-end of your application in python using *Psycopg2* and *Streamlit* library. Some useful pointers are below:
    - Psycopg2 official documentation: Psycopg – PostgreSQL database adapter for Python — Psycopg 2.9.1 documentation
    - Streamlit official documentation: Streamlit Docs
- The web-based application you develop in part 2 need to run locally. The instructor and TA and run your application locally using the code you submit.
- You should prepare slides for a presentation (~5 mins) describing your application.
- Each project partner should also prepare a statement of contributions: a document describing their own and their partner's contributions to the project. **This part of the assignment is to be completed individually.** Do not discuss the content of this document with your partner, and do not collaborate on this document.

Submission instructions

For part 2, submit your assignment through Canvas. Both project partners should submit the complete project by the deadline, with submissions that are identical except for the contributions statement. The submission should include a project report with no more than 4 pages, including updated description of the project, the business rules, the ER model, schema.sql and inserstion.sql, while sql statements need to be stored as single PDF files for execution. The schema.sql should include the creation of your database tables. The inserstion.sql should include all data and code that implements your data loading procedure. Finally, include the python file that implements the UI in your submission.

To submit through Canvas, upload a zip file with the name **<UTSAid1>_<UTSAid2>_project.zip** (e.g., abc123_xyz456_project.zip, with project partners' UTSA IDs in alphabetical order) with the following structure (code and data are folders):

- abc123_xyz456_project (zip archive)
    - **report.pdf** - project description, business rules, ER diagram
    - **slides.pdf** – presentation slides

- **contributions.pdf** - each project partner should draft this separately, explaining their own and their partner's contribution to the project.
- **code/**
  - **schema.sql** (SQL File)
  - **insertion.sql** (SQL File)
  - **project.py** (Python file)
  - …
- **data/**
  - **insertion.sql** (SQL File) with insert statements and/or data file(s) in CSV format
  - or code that implements your synthetic data creation procedure